

EDS Form Fragment Block — Behavior Specification

Executive Summary

The current AEM `form/xfinclusion` component includes best-match variation logic to serve the correct localized form fieldset to each visitor. This behavior is **fully preserved** in the EDS target architecture.

The mechanism changes — from server-side Java traversing the MSM hierarchy to client-side JavaScript reading the page URL — but the outcome is identical: the most specific matching variation is served, with a defined fallback chain ending at the global master.

Concern	Answer
Is best-match logic preserved?	Yes — fully preserved
Is it replaced?	The mechanism is replaced (Java → JS). The behavior is not.
What logic supports the target approach?	Defined in Section 3 of this document

1. Current AEM — Form XF Inclusion Component

What The Author Does

The author selects a fragment from a curated dropdown in the component dialog.

What Happens At Render Time

Build Locale Candidate List

For MSM live-copy pages:

```
1 Walk MSM hierarchy upward from current page
2 → Build ordered locale list: [lang-country, lang-region, lang]
3 → For each locale: check translation status (must be APPROVED or
  COMPLETE)
4 → If incomplete: remove that locale from list
5
```

For standalone (non-MSM) pages:

```
1 Parse locale from page URL path
2 → Build list: [lang-country, lang-region]
```

Match and Render

Walk the candidate list in order. Return the first variation that exists under the fragment. If none match, return `master`. Render the resolved variation with `wcmmode=disabled`.

2. EDS Target — Form Fragment Block

What It Is

A block placed on a form page that includes a localized form fieldset. It replaces `form/xfinclusion` in the EDS page model.

What The Author Does

The author enters one field — the base fragment path:

```
1 | form-fragment |
2 | ----- |
3 | /forms/fragments/standard/marketing-opt-in |
4 |
```

For a division-specific fragment:

```
1 | form-fragment |
2 | ----- |
3 | /forms/fragments/custom/lcd/marketing-opt-in |
4 |
```

No dropdown. No `fragmentType` field. The division is encoded in the path — Standard fragments use `standard/`, division fragments use `custom/{division}/`.

What The Block Does At Runtime

The block JavaScript reads the base path, determines the visitor's locale from the page URL, builds an ordered candidate list, and fetches the first variation that exists. Full logic is defined in Section 3.

3. Form Fragment Block — Execution Flow

```
1 PAGE LOADS - form-fragment block is on the page
2 |
3 ▼
4 READ base path from block
5 basePath = "/forms/fragments/standard/marketing-opt-in"
6 |
7 ▼
8 PARSE locale from page URL
9 URL: /us/en/home/forms/...
10 → country = "us"
```

```

11 → lang = "en"
12 |
13 ▼
14 BUILD ordered candidate list [most specific → least specific →
  master]
15 |
16 |— Always add: {lang}-{country}      → "en-us"
17 |
18 |— If lang ≠ en, also add: en-{country} → (e.g. "en-cn" for zh/cn
  page)
19 |
20 |— Look up country in region map:
21 |   us → "en-north-america"          → "en-north-america"
22 |
23 |
24 |— Always add: "master"              → "master"
25 candidates = ["en-us", "en-north-america", "master"]
26 |
27 ▼
28 FETCH candidate 1 → basePath/en-us.plain.html
29 |
30 |— 200 OK → use this variation → STOP
31 |— 404    → skip, try next
32 |
33 ▼
34 FETCH candidate 2 → basePath/en-north-america.plain.html
35 |
36 |— 200 OK → use this variation → STOP
37 |— 404    → skip, try next
38 |
39 ▼
40 FETCH candidate 3 → basePath/master.plain.html
41 |
42 |— 200 OK → use master (always exists) → STOP
43 |
44 ▼
45 RENDER matched variation HTML into block
46

```

Example — German Page

```

1 URL: /de/de/home/... → country=de, lang=de
2 Candidates:
3 1. "de-de"           (lang + country – German specific)
4 2. "en-de"           (lang ≠ en: English version for Germany)
5 3. "en-europe"       (region fallback from map)
6 4. "master"          (final fallback)
7 Fetch de-de         → 404 (no German variation for this form)
8 Fetch en-de         → 404
9 Fetch en-europe     → 200 OK ✓ → render European English variation
10

```

Example — Korean Page

```

1 URL: /kr/ko/home/... → country=kr, lang=ko
2 Candidates:
3 1. "ko-kr"           (Korean language, Korea country)
4 2. "en-kr"           (English version for Korea)
5 3. "en-ipac"         (IPAC region fallback)
6 4. "master"
7 Fetch ko-kr         → 200 OK ✓ → render Korean variation
8

```

4. What Is Preserved and What Is Simplified

What Is Preserved

Behavior	Status
Best-match variation selection	Preserved — most specific first, fallback to regional, fallback to master
Country-specific variations (en-us, en-ca, zh-cn, de-de, ko-kr ...)	Preserved
Regional fallback chain (en-north-america, en-europe, en-ipac ...)	Preserved
Non-English language handling (zh, de, ko, ja, es)	Preserved — tries native language first, then English equivalent, then region
Division-specific fragments (CUSTOM type)	Preserved — division encoded in path, author selects correct division path
Master as universal final fallback	Preserved
Renders nothing when path is wrong	Preserved — placeholder shown

What Is Simplified

1. Locale derivation — MSM hierarchy replaced by URL parsing

In AEM, locale was derived by walking the MSM live-copy tree upward from the current page.

In EDS, each locale URL is a distinct page. `/us/en/` is not the same page as `/de/de/`. The locale is already in the URL. URL parsing replaces MSM traversal entirely. The business outcome — correct locale is identified — is identical.

2. Translation status gate replaced by page existence

In AEM, each locale's translation status was checked (must be APPROVED or COMPLETE). If incomplete, that locale was removed from the candidate list, forcing fallback to the parent variation.

In EDS, publishing a fragment variation page is the equivalent of marking it ready. An unpublished or non-existent variation returns 404, which the block treats as "not ready" and skips — falling back to the next candidate. Same business outcome: unready content is never shown, parent variation is used instead.

3. **Logic runs in JS, not Java**

The best-match algorithm runs in the block's JavaScript in the browser instead of in a Sling Model on the server. The logic is equivalent — build candidate list, walk it, return first match. No behavioral change.

4. **Authoring — path field instead of dropdown**

In AEM, the author selected from a constrained dropdown populated by a datasource servlet.

In EDS, the author enters the fragment base path directly. The division is implicit in the path chosen. This is a change in authoring experience, not in runtime behavior. Author guidance can be implemented via a path browser in Universal Editor scoped to the fragments root.

5. **Direct Logic Mapping**

AEM Mechanism	EDS Equivalent
<code>fragmentPath</code> from author dialog dropdown	<code>path</code> field in form-fragment block — author enters base path
<code>fragmentType</code> STANDARD / CUSTOM hidden field	Encoded in path: <code>standard/</code> or <code>custom/{division}/</code>
MSM hierarchy walk → ordered locale list	URL path parsing → <code>[lang]-[country]</code> + region lookup
Translation status gate (APPROVED / COMPLETE)	Variation page existence — 404 = not ready = skip
<code>XFragmentInclusionService.getBestMatchedVariation()</code>	Ordered candidate list + sequential fetch — first 200 wins
Fallback to <code>master</code>	<code>master</code> always last in candidate list — always present

Render with `wcmmode=disabled`

Fetch with `.plain.html` endpoint
— content only, no editor chrome